

FieldFix: Grounded EV Field Copilot (RAG Engine)

Official Engineering Submission Report & Deliverables

Target System: Amperia Charging Networks (1,200 Commercial DC Fast Chargers: DC-150, DC-60, AC-22)
Core Architecture: Dual Sparse (Okapi BM25) + Dense (1,536d Cosine) + Reciprocal Rank Fusion (k=60) + Re-Ranker
Verification Watermark: SHA-256 Fingerprint: c6278cc1... | Pure Python Algorithms (Zero Framework Wrappers)

Core Question Answered: *How can an application answer questions from private, current source material that the model was never trained on, without sending the entire source collection to the model for every question?*
Solution: FieldFix combines exact inverted-index token matching (Okapi BM25) for rare error codes (E-217, E-104) and dense semantic vector search (text-embedding-3-small) for colloquial technician phrasing, fusing ranks via Reciprocal Rank Fusion (RRF, k=60) to inject only top-k evidence into a strictly grounded LLM prompt with chunk citations and refusal directives.

1. Chunking Strategy & Production Sizing (Deliverable 1)

We benchmarked three distinct chunking algorithms at small (40w) and production (200w) thresholds across the 3,984-word Amperia corpus:

Document	Fixed (40w)	Fixed (200w)	Structure (40w)	Structure (200w) [Selected]	Semantic (40w)	Semantic (200w)
dc150_product_manual	28	6	25	9	31	21
firmware_release_notes	23	5	25	14	27	16
incident_postmortem	24	5	21	7	29	16
operator_sla_pricing	17	4	16	8	19	12
site_inspection (parsed & pdf)	30	7	25	7	32	18
technician_shift_notes	11	2	9	2	16	10
TOTAL CORPUS CHUNKS	133	29	121	47 Units	154	93

Production Choice & Evidence: We selected the **Recursive Structure-Aware Chunker (max 200 words, min 25 words)**. Fixed-size chunking suffered from fact-splitting: at 40 words, it split the definition of E-217 into Chunk #24 while severing the vital instruction (*'Classification: Firmware Timing Logic. NOT a hardware defect. Do NOT replace contactors'*) into Chunk #25. If retrieved alone, the copilot recommends an erroneous \$7,800 contactor swap. The Structure-Aware chunker preserved the entire error definition and operational prohibition together in chunk struct_200w_c008 (173 words).

2. Head-to-Head: Where BM25 Won vs. Where Vector Search Won (Deliverable 2)

- BM25 Won on Exact Alphanumerics ('E-217'):** The token 'e-217' has high Inverse Document Frequency (IDF $\approx$ 2.26). BM25 ranked the Known Field Advisory #FWA-2026-03 and Error Table at **Rank #1 and #2** with zero noise (score: 6.25). Vector search dispersed similarity across general error sections because alphanumeric strings lack distinct natural language semantic structure.
- Vector Search Won on Colloquial Phrasing ('how long is the power module covered if it dies?'):** The technician used informal terms (*'covered', 'dies'*), whereas the manual uses formal terms (*'Warranty Schedule', '5 Years', 'thermal fatigue'*). BM25 scored 0.0 for the warranty chunk due to vocabulary mismatch. Vector search mapped the conceptual intent, placing the 5-year warranty table at **Rank #1** (cosine: 0.5251).

3. Hybrid Search (RRF) & Re-Ranking Scorecard Impact (Deliverable 3)

Strategy	Recall @3	Recall @5	Latency	Primary Strength	Primary Weakness
Okapi BM25	25/26 (96.2%)	26/26 (100.0%)	~2 ms	Pinpoint exact error codes & part numbers	Vocabulary mismatch on colloquial queries
Dense Vector (1536d)	24/26 (92.3%)	25/26 (96.2%)	~45 ms	Conceptual intent & paraphrased questions	Dilutes alphanumeric codes & torque numbers

Hybrid (RRF k=60)	25/26 (96.2%)	25/26 (96.2%)	~48 ms	Best baseline recall; zero parameter tuning	Slightly larger candidate payload
Hybrid + Re-Rank	26/26 (100.0%)	26/26 (100.0%)	~280 ms	100% Top-3 Precision; actionable fixes at #1	Re-ranking compute & latency overhead

Why Adding Raw BM25 to Cosine Scores Fails: BM25 scores are unbounded positive numbers (0 to 25+) scaling with query length, while cosine similarities are bounded in [-1, 1]. Naive addition lets BM25 swamp cosine similarity by 10× to 30×, completely drowning out semantic matching. Reciprocal Rank Fusion (RRF, $score = \sum 1 / (60 + rank)$) resolves this by fusing ordinal ranks.

Was it Worth the Complexity? Hybrid RRF is **100% worth it** (~40 lines of code, +3 ms latency, eliminates single-retriever failure). Re-ranking is **essential for safety-critical diagnostics**, elevating Top-3 recall to 100% and surfacing direct rollback instructions.

4. Failure Mode Analysis & Engineering Remediation (Deliverable 4)

- **Retrieval Failure:** Query: *'How long is the power module covered if it fails?'* Informal keyword *'fails'* biased BM25 toward error logs, dropping the warranty chunk (c005) to Rank #11 (outside Top-10). The grounded reader correctly refused (*'I don't have enough information'*). **Fix:** Query expansion (mapping *'covered'* → *'warranty schedule'*), HyDE, and expanding candidate depth from k=10 to k=25.
- **Generation Failure:** Query: *'Total parts expenditure incurred for the Pune repairs?'* Both post-mortem and shift notes were present in prompt context. Standard LLMs suffer from numerical salience and temporal blindness, latching onto *'\$7,800 estimated parts expense'* at 01:30 IST and missing the 02:45 IST resolution (*'emergency RMA cancelled; parts returned; \$0 incurred'*). **Fix:** Chain-of-Thought timeline verification prompting and cross-chunk reconciliation.

5. Re-Ingestion, Blue-Green Versioning & Zero-Downtime Rollback (Deliverable 5)

- FieldFix implements an **Immutable Blue-Green Vector Store** in data/vector_store.json:
- **Corpus Watermarking:** Deterministic SHA-256 fingerprint (v4.3: c6278cc1..., v4.4: b586b73e...).
 - **Non-Destructive Coexistence:** Re-ingesting v4.4.0 adds 47 new chunks while retaining the 42 prior chunks (89 total chunks). Queries filter strictly on chunk.corpus_id == active_corpus_id.
 - **Instant Rollback Runbook:** Executing store.rollback_to('c6278cc1...') switches the active version pointer in **< 1 ms** at **\$0.00 cost** (zero re-embedding calls). The pipeline immediately refuses queries regarding unvetted v4.4.0 features, proving complete version isolation.

6. Observability, Cost at 10,000 Questions/Day & #1 Optimization (Deliverable 6)

- Every request logs stage latencies (embed, retrieve, rerank, generate), token counts, and cost via unique correlation IDs (req_20261003_5d1c9907):
- **Measured Per-Request Cost:** 1,206 prompt tokens (\$0.15/M) + 1,442 completion tokens (\$0.60/M) + 30 embed tokens (\$0.02/M) = **\$0.001222 USD / request**.
 - **Cost at 10,000 Questions per Day:** **\$12.22 / day | \$366.60 / month | \$4,460.30 / year**.
 - **#1 Change That Reduces Cost Most: Semantic & Exact Response Caching (Redis).** In field engineering, 65%–75% of technician queries are repeat lookups for the same top-20 error codes (E-217, E-104) and torque specs (55 Nm). A semantic cache with cosine similarity > 0.96 bypasses LLM inference entirely for cached hits (\$0.00 cost, < 5 ms latency). At a **70% cache hit rate**, daily LLM calls drop to 3,000, reducing daily spend from **\$12.22 to \$3.67 / day** — a **70% net fleet operational savings!**

Compliance & AI Usage Statement: All core algorithms (BM25, RRF, Chunkers, Eval) were implemented from scratch. AI tools were used solely for boilerplate CLI formatting and synthetic PDF generation. Repository verified to run cleanly from clone.